

Project Workflow: Week 6

Phase: Ball Physics & Collision Detection

1. Objective

The primary goal of Week 6 was to implement the ball movement and collision system to make the Ping Pong game interactive and playable. This included:

- Implementing continuous ball movement.
- Creating a Timer-based game loop.
- Detecting collisions between the ball and paddles.
- Detecting collisions between the ball and game boundaries.
- Creating smooth ball bouncing behavior.

2. Detailed Task Breakdown

- Ball Movement: Implemented continuous movement of the ball using X and Y coordinates.
- Game Loop: Used the Java Timer class to update the ball position continuously and provide smooth gameplay.
- Paddle Collision: Added collision detection between the ball and both paddles so the ball changes direction after hitting a paddle.
- Wall Collision: Implemented collision detection with the top and bottom boundaries of the game area.
- Physics Testing: Tested different ball positions and movement directions to ensure that the ball bounces correctly and does not pass through the paddles or walls.

3. Deliverables

- Functional moving ball.
- Timer-based game loop.
- Paddle collision detection.
- Wall collision detection.
- Smooth ball bouncing system.
- Tested basic gameplay mechanics.

4. Tools & Resources

- Programming Language: Java
- GUI Framework: Java Swing
- Game Loop: Java Timer
- Collision Handling: Java AWT / Rectangle
- Development Environment: IntelliJ IDEA / VS Code
- Reference Materials: Java Swing, AWT, Timer, and collision detection references